Computer Vision Internship Assessment - Report

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1. Overview: Purpose and Deliverables The goal of this assessment was to demonstrate foundational skills in computer vision using Python and OpenCV through two practical tasks. Task 1: Basic Image Manipulation

- Developed a script to perform image processing operations: grayscale conversion, Gaussian blur, and Canny edge detection.
- Used OpenCV functions to process an input image and saved the results (grayscale, blurred, edges) as separate files.

Task 2: Real-Time Color Detection

- Built a webcam-based application to identify pixel colors on click using OpenCV's cv2.setMouseCallback.
- Displayed BGR color values of the selected pixel in real time.
- Integrated a color dataset (colors.csv) to identify the closest named color using Euclidean distance in RGB space. The detected color name and values were shown on the video feed.

2. Challenges Faced

- Environment Setup: Ensuring compatible Python and OpenCV versions.
- Image Processing Parameters: Fine-tuning kernel sizes and threshold values for optimal visual output.
- Mouse Callback Handling: Passing the correct frame to access accurate pixel values during clicks.
- Color Matching Logic: Sourcing a clean dataset and calculating distances efficiently; recognizing RGB limitations.
- Real-Time Updates: Maintaining smooth performance, positioning overlays, and avoiding lag.

3. Suggested Improvements

- **Task 1 Enhancements:** Add command-line options, parameter customization, and input validation.
- **Task 2 Enhancements:**
- Use LAB color space for better perceptual accuracy.
- Show color details near the cursor with a larger swatch.
- Add trackbars for real-time parameter tuning.
- Optimize large dataset search with structures like k-d trees.
- Add robust error handling (e.g., webcam disconnection).
- Average over pixel neighborhoods to reduce noise.